Practical 2a: PWA with Offline Mode using Service Worker

AIM : To develop a PWA that works in offline mode using Service Worker with cache-first strategy.

service-worker.js (Cache First Strategy)

```
const CACHE_VERSION = 'v2';
const CACHE_NAME = `pwa-cache-${CACHE_VERSION}`;

const STATIC_ASSETS = [
  '/',
  '/index.html',
  '/style.css',
  '/app.js',
  '/icons/icon-192.png',
  '/icons/icon-512.png',
  '/offline.html'
];

// Install: Cache all static assets
self.addEventListener('install', event => {
  console.log('[SW] Install');
  event.waitUntil(
    caches.open(CACHE_NAME)
      .then(cache => cache.addAll(STATIC_ASSETS))
      .then(() => self.skipWaiting())
  );
});

// Activate: Clean old caches
self.addEventListener('activate', event => {
  console.log('[SW] Activate');
  event.waitUntil(
    caches.keys().then(keys =>
      Promise.all(
        keys.filter(k => k !== CACHE_NAME).map(k => caches.delete(k))
      )
```

```
    ).then(() => self.clients.claim())
  );
});

// Fetch: Cache First, Network Fallback
self.addEventListener('fetch', event => {
  event.respondWith(
    caches.match(event.request).then(cached => {
      if (cached) {
        console.log('[SW] Serving from cache:', event.request.url);
        return cached;
      }
      return fetch(event.request).then(response => {
        const clone = response.clone();
        caches.open(CACHE_NAME).then(cache => cache.put(event.request, clone));
        return response;
      }).catch(() => caches.match('/offline.html'));
    })
  );
});
```

Output:

